Machine Learning and Forecasting in Atlassian App Analytics

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Contents

1 Introduction

This document describes the two most valuable analytical components of the Atlassian App Analytics platform:

1. Machine Learning Churn Prediction
2. Trend-Based Forecasting

Together, these components transform the application from a reporting dashboard into a predictive decision-support system.

2 Machine Learning Churn Prediction

2.1 Business Objective

The machine learning model estimates the probability that a customer will churn.

Typical outputs include:

- 12% churn probability (low risk)
- 47% churn probability (moderate risk)
- 86% churn probability (critical risk)

This enables customer success and product teams to prioritize interventions.

2.2 Training Features

The model is trained using the following variables:

Feature	Description
usage_score	Frequency of product activity
feature_adoption_score	Breadth of feature usage
reliability_score	Product stability experienced
support_score	Support burden
company_size	Size of the customer organization

2.3 Target Variable

The target variable is:

```
did_churn = health_score < 40
```

This synthetic label allows model training even when actual churn labels are unavailable.

2.4 Model Choice

The chosen algorithm is Logistic Regression because it:

- Produces probabilities
- Is interpretable
- Trains quickly
- Performs well on structured data

2.5 Mathematical Foundation

The model estimates:

$$P(\text{churn}) = \frac{1}{1 + e^{-z}}$$

where

$$z = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

2.6 Training Process

1. Extract features and target labels from the database.
2. Split data into training and testing sets.
3. Train the model.
4. Evaluate accuracy.
5. Save the model to disk.

Example training code:

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
accuracy = accuracy_score(y_test, model.predict(X_test))
joblib.dump(model, MODEL_PATH)
```

2.7 Prediction Process

When scoring a customer:

1. Load the saved model.
2. Build the feature vector.
3. Call `predict_proba()`.
4. Extract the probability of churn.

```
probability = model.predict_proba(features)[0][1]
```

If the model returns:

```
[0.18, 0.82]
```

the customer receives an 82% churn probability.

2.8 Feature Importance

Logistic regression coefficients reveal the drivers of churn.

- Positive coefficient = increases churn risk
- Negative coefficient = decreases churn risk
- Larger absolute value = stronger influence

2.9 Historical Preservation

Each `CustomerHealthSnapshot` stores the current:

- health_score
- churn_risk
- ml_churn_probability

This preserves exactly what the model predicted at that moment.

2.10 Business Questions Answered

Machine learning answers:

- Which customers are most likely to churn?
- How certain is the prediction?
- Which factors drive risk?
- How does ML compare to rule-based scoring?

3 Trend-Based Forecasting

3.1 Business Objective

The forecasting engine estimates:

> What is the customer's likely health score next period if the current trend continues?

3.2 Data Source

Forecasting uses `CustomerHealthSnapshot` records.

Example history:

$$80 \rightarrow 75 \rightarrow 70 \rightarrow 65 \rightarrow 60$$

3.3 Forecasting Algorithm

The function:

```
forecast_next_health_score(customer)
```

performs the following steps:

1. Retrieve up to five recent snapshots.
2. Compute total score change.
3. Divide by the number of intervals.
4. Calculate average change.
5. Add average change to the latest score.
6. Clamp the result to the range 0 to 100.

3.4 Mathematical Formula

$$\text{Average Change} = \frac{\text{Latest Score} - \text{First Score}}{n - 1}$$

$$\text{Forecast} = \text{Latest Score} + \text{Average Change}$$

3.5 Worked Example

Historical scores:

$$80 \to 75 \to 70 \to 65 \to 60$$

Total change:

$$60 - 80 = -20$$

Intervals:

$$4$$

Average change:

$$-20/4 = -5$$

Forecast:

$$60 + (-5) = 55$$

Predicted next health score: **55**

3.6 Interpretation

The forecast means:

> If recent deterioration continues, the next health score will likely be 55.

3.7 Business Questions Answered

Forecasting answers:

- What may happen next?
- Is the customer improving or deteriorating?
- How urgently should we intervene?

4 Combined Predictive Intelligence

Machine learning and forecasting answer different but complementary questions.

Question	Capability
What is the churn probability right now?	Machine Learning
What is likely to happen next?	Forecasting
Why is risk increasing?	Feature Importance
What should we do?	Recommendations

4.1 Example Interpretation

Metric	Value
Current Health Score	58
ML Churn Probability	74%
Forecasted Next Health Score	52

Interpretation:

- The customer is currently high risk.
- Machine learning predicts a strong likelihood of churn.
- Historical trends indicate further deterioration.
- Immediate intervention is recommended.

5 Strategic Value

These analytical components enable organizations to:

1. Measure customer health
2. Predict churn probability
3. Detect worsening trends

4. Forecast future deterioration
5. Explain risk drivers
6. Recommend corrective actions

6 Conclusion

The machine learning and forecasting modules are the analytical core of Atlassian App Analytics.

The machine learning model quantifies the current probability of churn, while the forecasting engine projects future customer health based on historical trends.

Together, they enable proactive customer retention rather than reactive reporting.

Useful Links

- GitHub: https://github.com/karianjahi/atlassian-app-analytics
- Live Demo: https://customer-health-app.onrender.com
- Contact: mailto:nkarianjahi@gmail.com